

IS 3920 – Individual Project on Business Solutions

Software Requirements Specification

for

Online Tuition Class Management Platform

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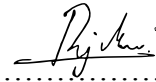
15/01/2026

Student's Declaration

I hereby declare that this report is my original work and has not been submitted, in whole or in part, for any degree or diploma at any university or other institution of higher learning. All information derived from the work of others, whether published or unpublished, has been properly acknowledged in the text, and a complete list of references has been provided.

18/01/2026

Date



Signature of Student

Supervisors' declaration

I hereby declare that I have reviewed this project and find it adequate in both scope and quality.

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Any further comments:

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Revision History

Name	Date	Reason for Changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the functional and non-functional requirements of the Online Tuition Class Management Platform. This document acts as a formal agreement between stakeholders, including developers, supervisors, academic evaluators, and potential users, regarding what the system is expected to do and how it should perform.

The SRS serves as a foundational reference throughout the system development life cycle. It supports requirement validation, system design, implementation, testing, and final evaluation. By clearly defining system behavior, constraints, assumptions, and requirements at an early stage, the document minimizes ambiguities and reduces the risk of misunderstandings during development.

Additionally, this document is prepared to fulfill the academic requirements of the IS 3920 - Individual Project on Business Solutions module and follows standard software engineering documentation practices.

1.2 Document Conventions

This document adheres to IEEE Software Requirements Specification standards. Standardized section numbering, terminology, and formatting are used to ensure clarity and consistency. Functional requirements are expressed using mandatory language such as “shall” to clearly indicate system obligations.

Headings and subheadings are structured hierarchically to improve readability. Bullet points are used where appropriate to present information concisely, while explanatory paragraphs are included to elaborate on key concepts and system behaviors. Technical terms and abbreviations are explained in the glossary section.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Project supervisors and mentors
- Software developers
- System evaluators
- Academic examiners

Readers are advised to first review the Introduction and Overall Description sections to understand system scope, followed by System Features and Nonfunctional Requirements for detailed specifications.

1.4 Product Scope

The Online Tuition Class Management Platform is a centralized, web-based system designed to support the end-to-end management of online tuition classes. The platform brings together essential features such as student registration, online fee payment, resource sharing, teacher–student communication, and teacher evaluation into a single integrated environment.

The system aims to simplify administrative tasks for tuition providers while enhancing the learning experience for students and teaching experience for educators. By eliminating the need for multiple disconnected tools, the platform improves efficiency, transparency, and usability.

1.4.1 Problem in brief

Many online tuition institutes currently rely on a combination of informal and fragmented tools to manage their operations. Communication is often handled through messaging applications such as WhatsApp, payments are processed manually via bank transfers, and study materials are shared using external file-sharing services.

These practices lead to several challenges, including lack of centralized records, difficulty in tracking payments, inconsistent communication, limited accountability, and poor overall user experience. Students may miss important updates, teachers may struggle to manage resources effectively, and administrators may face difficulties in monitoring activities.

1.4.2 Aim and Objectives

Aim:

- The primary aim of this project is to develop a secure, scalable, and user-friendly web-based tuition management platform that integrates essential online class management functionalities into a single system.

Objectives:

- To design and develop an intuitive user interface that supports students, teachers, and administrators.
- To implement secure online payment processing for tuition fees.
- To enable teachers to upload and manage study materials and online class links.
- To provide secure authentication and role-based access control.
- To introduce a transparent teacher rating and review mechanism.
- To deploy the system on a cloud platform to ensure availability and scalability.

1.5 References

- IEEE Std 830-1998 – Software Requirements Specification.
- Academic literature related to online learning management systems.
- Official documentation of Stripe and PayHere payment gateways.

2. Overall description

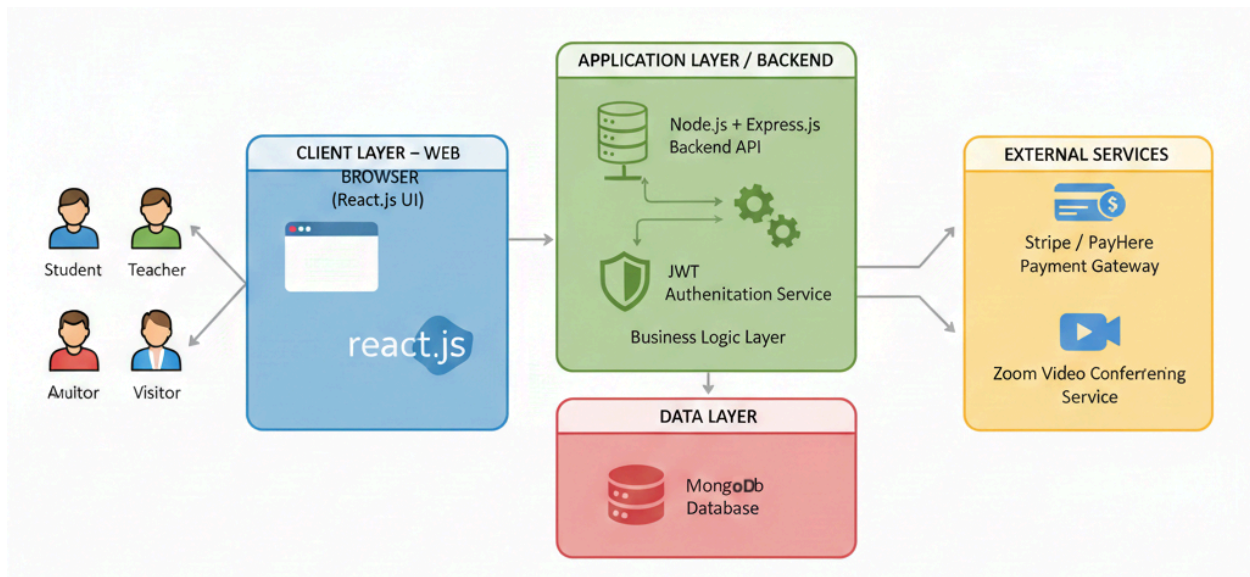


Figure 1 - System Architecture

2.1 Product Perspective

The Online Tuition Class Management Platform is designed as a standalone, cloud-based web application that supports online tuition class operations. The system follows a client–server architecture where users interact with the application through a web browser, while all business logic and data processing are handled on the server side.

The frontend of the system is developed using React.js, which provides a responsive and interactive user experience. The backend is implemented using Node.js with the Express.js framework, which manages API requests, authentication, and business logic. MongoDB is used as the database to store user data, class details, payment records, uploaded resources, and teacher ratings.

The platform integrates with external third-party services such as payment gateways (Stripe and PayHere) for secure online transactions and video conferencing platforms (e.g., Zoom) for conducting live online classes. These integrations allow the system to function as a complete solution rather than relying on multiple disconnected tools.

The system is intended to be deployed on a cloud hosting environment, ensuring scalability, availability, and accessibility from any location. It is designed to support growth in the number of users, classes, and transactions without requiring major architectural changes.

2.2 Product Functions

The major functions of the Online Tuition Class Management Platform include, but are not limited to, the following:

- **User Registration and Authentication:** Allows students, teachers, and administrators to create accounts and securely log in to the system.
- **Role-Based Access Control:** Ensures that users can only access features relevant to their assigned roles.
- **Teacher Management:** Enables teachers to register, submit profile details, and await administrative approval.
- **Student Management:** Allows students to register for classes and manage their enrolled courses.
- **Online Payment Processing:** Integrates secure payment gateways to handle tuition fee payments.
- **Resource Management:** Allows teachers to upload study materials such as documents, links, and announcements.
- **Class Link Management:** Enables teachers to share Zoom or online class links with enrolled students.
- **Teacher Rating and Review System:** Allows students to rate and review teachers based on their learning experience.
- **Administrative Monitoring:** Enables administrators to monitor users, payments, content, and system usage.
- **Reporting and Analytics:** Generates basic reports related to payments, user activity, and class engagement.

2.3 User Classes and Characteristics

Admin:

- The administrator is responsible for managing the overall system. Admin users have advanced privileges, including approving teacher registrations, managing user accounts, monitoring financial transactions, and ensuring content quality. Admin users are expected to have good technical and administrative knowledge.

Teachers:

- Teachers are responsible for conducting online tuition classes. They can upload study materials, share online class links, post announcements, and view student feedback. Teachers require basic digital literacy and familiarity with online teaching tools.

Students:

- Students use the platform to register for classes, make online payments, access learning resources, and participate in online sessions. Students are expected to have basic internet and computer skills.

Visitors:

- Visitors can browse publicly available information such as teacher profiles, ratings, and available classes without registering. This user class helps potential students make informed decisions.

2.4 Operating Environment

The system operates in the following environment:

- Client Side: Any modern web browser such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.
- Server Side: Cloud-based Node.js runtime environment.
- Database: MongoDB NoSQL database.
- Network: Internet connection using secure HTTPS protocol.
- External Services: Stripe/PayHere payment gateways and Zoom or similar video conferencing platforms.

2.5 Design and Implementation Constraints

The development and implementation of the system are subject to the following constraints:

- Limited project duration as defined by academic timelines.
- Use of JavaScript-based technologies as per project scope.
- Dependence on third-party services for payments and video conferencing.
- Compliance with data protection and privacy regulations.
- Budget constraints that limit the use of paid third-party tools.

2.6 User Documentation

The following user documentation will be provided:

- Administrator User Manual describing system configuration and monitoring features.
- Teacher User Guide explaining class management, resource uploads, and communication tools.
- Student User Guide detailing registration, payment, and resource access processes.
- Online help pages and FAQs for quick reference.

2.7 Assumptions and Dependencies

Assumptions:

- Users have access to a stable internet connection.
- Users possess basic digital literacy.
- Payment gateway services remain operational.

Dependencies:

- Availability and reliability of cloud hosting services.
- Continued support from third-party payment and video conferencing services.

3. External Interface Requirements

3.1 User Interfaces

- The Online Tuition Class Management Platform provides a responsive, web-based user interface designed according to modern UI/UX principles. The interfaces are role-based and optimized for ease of use, accessibility, and clarity. The system follows consistency in layout, color schemes, and navigation across all user roles. UI designs (wireframes or mockups) related to each interface will be included under the relevant subsections.

3.1.1 Visitor Interface (Public View)

3.1.1 Visitor Interface (Public View)

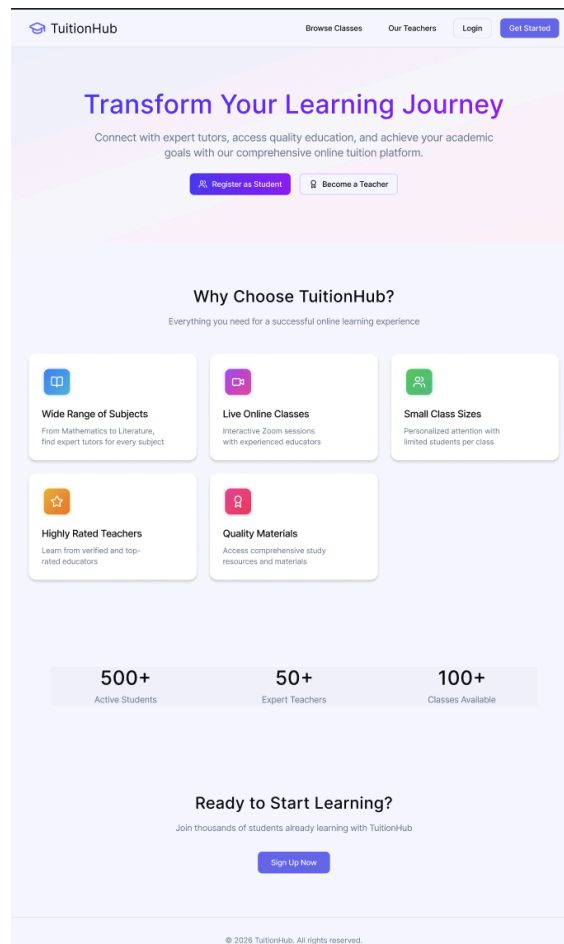


figure 2-Visitor Interface (Public View)

Description: This interface is accessible without user authentication and serves as the entry point for new users. It is designed to provide an overview of the platform and help visitors make informed decisions.

Key UI Components:

- Landing page with platform introduction and key features.
- Navigation bar with options to Login, Register, and Browse Teachers.
- Teacher listing view displaying subject, experience, and average rating.
- Teacher rating overview with star ratings and student feedback summaries.

Purpose: To allow visitors to explore available classes and teachers before registration.

3.1.2 Authentication Interfaces (Login & Registration)

The figure displays three screenshots of the TuitionHub authentication interfaces:

- Login Interface:** Features the TuitionHub logo, a "Welcome Back" message, and a "Sign in to your account" prompt. It includes a "Login" section with role selection (Student, Teacher, Admin), email and password input fields, a "Sign In" button, and a "Register here" link for new users.
- Create Account Interface:** Features the TuitionHub logo, a "Create Account" title, and a "Join our learning community today" prompt. It includes a "Registration" section with role selection (Student, Teacher), full name and email input fields, password and confirm password input fields, a "Create Account" button, and a "Login here" link for existing users.
- Detailed Registration Form:** Shows the "Registration" section with a "Fill in your details to get started" prompt. It includes a "Teacher Information" section with "Subject" and "Years of Experience" input fields, a "Qualification" input field, and a "Create Account" button. A "Please fill in all required fields" message is displayed below the form.

figure 3-Authentication Interfaces (Login & Registration)

Description: These interfaces allow users to securely access the system based on their roles.

Key UI Components:

- Login screen for Students, Teachers, and Admins.
- Registration form for Students.
- Registration form for Teachers with profile and subject details.
- Role selection during login.
- Validation messages and error handling.

Purpose: To ensure secure and role-based access to the system.

3.1.3 Student Dashboard Interface

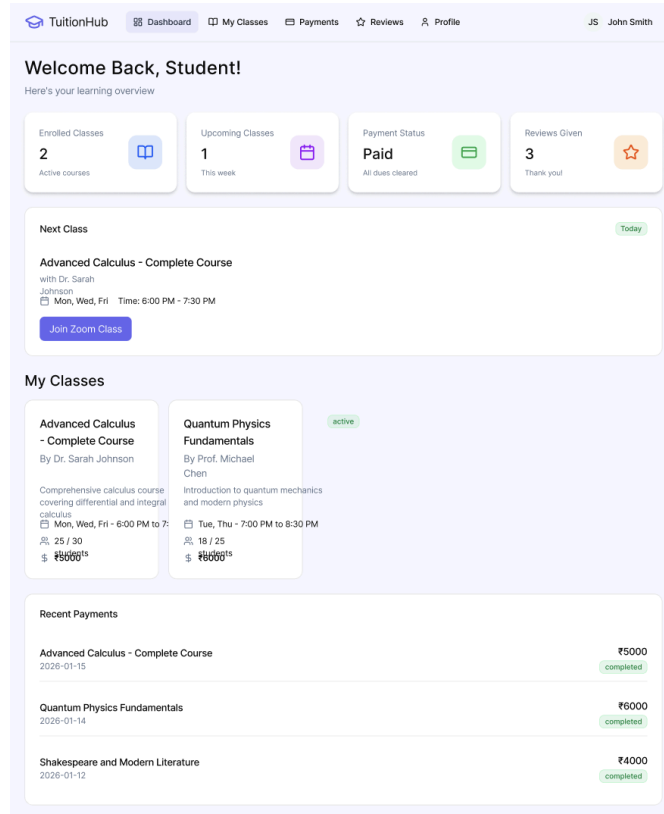


figure 4-Student Dashboard Interface

Description: The Student Dashboard acts as the main workspace for enrolled students.

Key UI Components:

- Overview panel displaying enrolled classes.
- Payment status indicators.
- Access links for Zoom/online classes.
- Study materials download section.
- Teacher rating and review submission interface.
- Notification and announcements panel.

Purpose: To provide students with centralized access to learning resources and class information.

3.1.4 Teacher Dashboard Interface

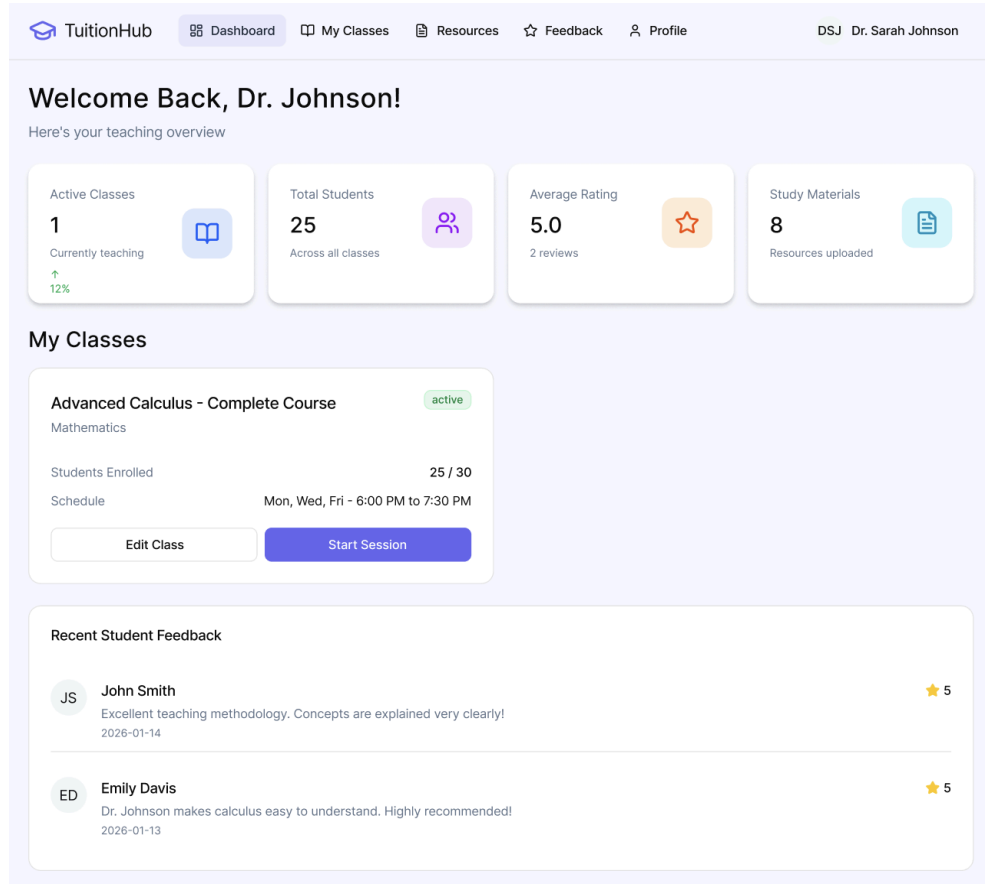


figure 5-Teacher Dashboard Interface

Description: The Teacher Dashboard enables teachers to manage their online tuition activities.

Key UI Components:

- Class management panel.
- Study material upload section.
- Zoom/online class link sharing interface.
- Student list and enrollment view.
- Ratings and feedback visualization.
- Announcement publishing section.

Purpose: To allow teachers to efficiently manage classes and interact with students.

3.1.5 Admin Dashboard Interface

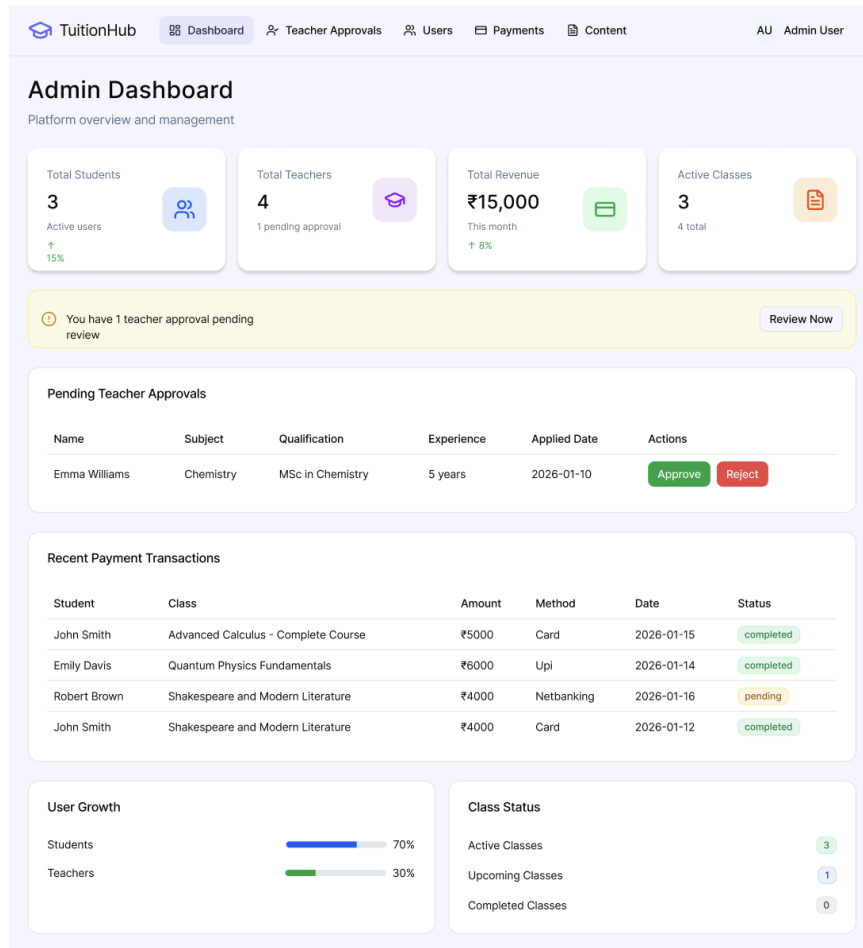


figure 6-Admin Dashboard Interface

Description: The Admin Dashboard is a management-oriented interface designed for system administrators.

Key UI Components:

- User management tables (students and teachers).
- Teacher approval and verification panel.
- Payment monitoring and reports view.
- Content moderation tools.
- System activity overview and analytics

Purpose: To enable administrators to monitor, control, and maintain system operations.

3.1.6 Teacher Rating and Review Interface

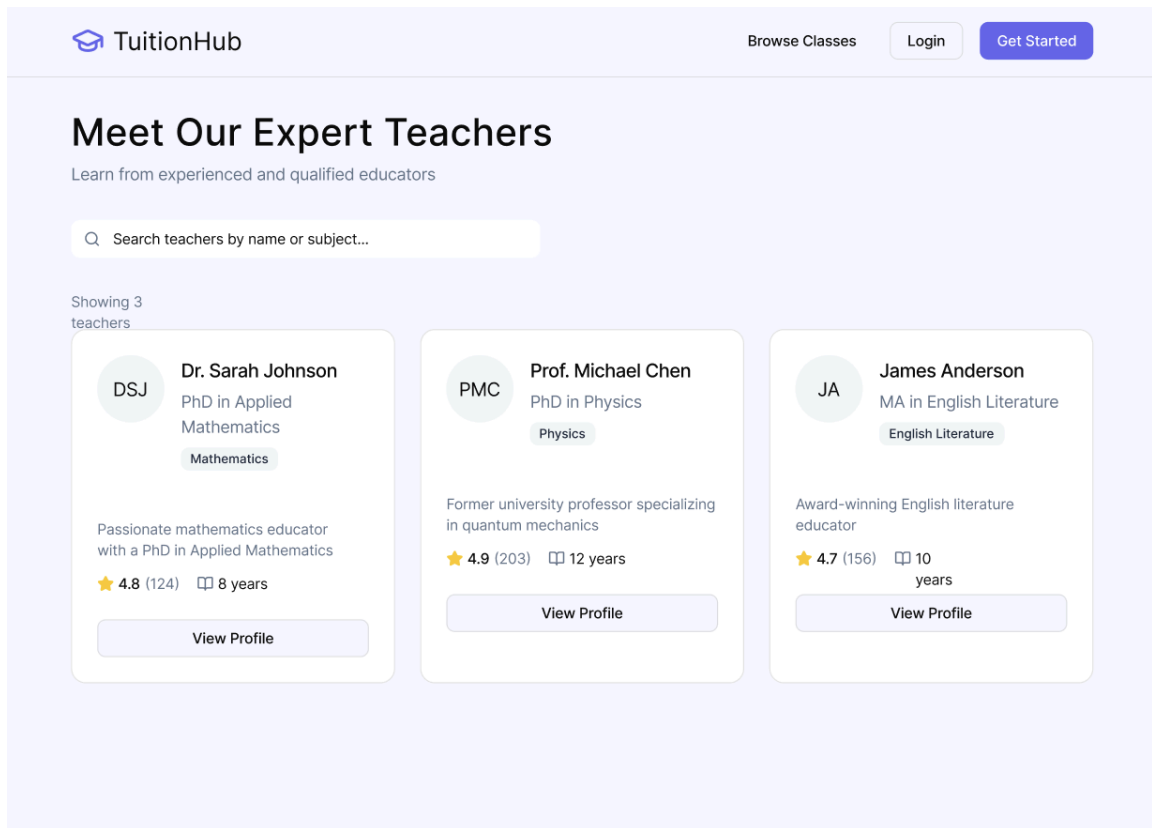


figure 7-Teacher Rating and Review Interface

Description: This interface allows students to submit feedback and ratings for teachers after class participation.

Key UI Components:

- Star-based rating input.
- Text-based review form.
- Review submission confirmation.
- Aggregated rating display on teacher profiles

Purpose: To promote transparency and quality assurance within the platform.

3.2 Hardware Interfaces

- No direct hardware interfaces are required. The system operates on standard computing devices.

3.3 Software Interfaces

- Payment Gateway APIs (Stripe / PayHere)
- Authentication libraries (JWT)
- Database interfaces (MongoDB drivers)

3.4 Communication Interfaces

- HTTPS for secure data transmission.
- RESTful APIs for client–server communication.
- Encrypted authentication tokens.

4. System Features

4.1 User Registration Feature

Description: This feature allows new users to create accounts on the platform as Students or Teachers. It captures essential personal information and role-specific details and stores them securely in the system.

Priority: High

Stimulus / Response Sequence:

- A visitor selects the Register option.
- The system displays a role-based registration form.
- The user enters required details and submits the form.
- The system validates the inputs and creates a user account.

Functional Requirements:

- The system shall allow students to register using name, email address, and password.
- The system shall allow teachers to register by providing subject areas and qualifications.
- The system shall validate mandatory fields before accepting registration.
- The system shall prevent duplicate registrations using the same email address.
- The system shall store registration data securely in the database.

4.2 User Authentication and Authorization Feature

Description: This feature enables registered users to log in to the system and access functionalities based on their assigned roles (Student, Teacher, or Admin).

Priority: High

Stimulus / Response Sequence:

- A user enters login credentials.
- The system validates the credentials.
- The system grants access according to the user role.

Functional Requirements:

- The system shall authenticate users using email and encrypted passwords.
- The system shall restrict access to features based on user roles.
- The system shall generate secure authentication tokens after successful login.
- The system shall display error messages for invalid login attempts.

4.3 Teacher Approval and Verification Feature

Description: This feature allows administrators to review and approve teacher registrations before granting full access to teaching functionalities.

Priority: High

Functional Requirements:

- The system shall allow administrators to view pending teacher registrations.
- The system shall allow administrators to approve or reject teacher accounts.
- The system shall notify teachers regarding approval status.
- The system shall restrict unapproved teachers from creating classes.

4.4 Student Dashboard Management Feature

Description: This feature provides students with a personalized dashboard that displays enrolled classes, payment status, study materials, and notifications.

Priority: High

Functional Requirements:

- The system shall display a list of enrolled classes for each student.
- The system shall allow students to access online class links.
- The system shall allow students to download available study materials.
- The system shall display notifications and announcements.

4.5 Teacher Dashboard Management Feature

Description: This feature enables teachers to manage their classes, upload learning materials, communicate with students, and view feedback.

Priority: High

Functional Requirements:

- The system shall allow teachers to create and manage classes.
- The system shall allow teachers to upload and update study materials.
- The system shall allow teachers to share online class links.
- The system shall allow teachers to post announcements to students.
- The system shall display student ratings and feedback to teachers.

4.6 Online Payment Processing Feature

Description: This feature enables secure online payment processing for tuition fees through integrated payment gateways.

Priority: High

Functional Requirements:

- The system shall integrate with approved online payment gateways.
- The system shall process tuition fee payments securely.
- The system shall confirm payment success or failure.
- The system shall grant class access only after successful payment.
- The system shall store payment transaction records.

4.7 Study Material Management Feature

Description: This feature allows teachers to upload learning resources while enabling students to access materials for enrolled classes.

Priority: Medium

Functional Requirements:

- The system shall allow teachers to upload documents and links.
- The system shall restrict access to materials to enrolled students only.
- The system shall allow students to download study materials.

4.8 Online Class Link Management Feature

Description: This feature allows teachers to share live online class links with enrolled students.

Priority: Medium

Functional Requirements:

- The system shall allow teachers to add and update online class links.
- The system shall display class links only to enrolled students.
- The system shall notify students when new class links are added.

4.9 Teacher Rating and Review Feature

Description: This feature enables students to submit ratings and reviews for teachers after attending classes.

Priority: Medium

Functional Requirements:

- The system shall allow students to submit ratings using a star-based system.
- The system shall allow students to submit textual reviews.
- The system shall calculate and display average teacher ratings.
- The system shall display ratings on public teacher profiles.

4.10 Admin Monitoring and Reporting Feature

Description: This feature provides administrators with tools to monitor system usage, users, and transactions.

Priority: Medium

Functional Requirements:

- The system shall allow administrators to view registered users.
- The system shall allow administrators to monitor payment transactions.
- The system shall generate basic system usage reports.
- The system shall allow administrators to manage system content.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Page response time shall be under 3 seconds.
- System availability shall be at least 99%.

5.2 Safety Requirements

- Secure payment handling.
- Data backup and recovery mechanisms.

5.3 Security Requirements

Security is paramount as the system handles financial data and personal information.

- Authentication: The system shall utilize JSON Web Tokens (JWT) for session management. Tokens shall expire after 2 hours of inactivity to prevent session hijacking.
- Encryption: All passwords shall be hashed using a strong algorithm (e.g., BCrypt) before being stored in MongoDB.
- Data Integrity: All API requests between the React frontend and Node.js backend must be conducted over HTTPS to ensure data is encrypted in transit.

5.4 Software Quality Attributes

- Usability.
- Reliability.
- Maintainability.
- Scalability.

5.5 Business Rules

- Only approved teachers can publish classes.
- Payments are mandatory for material access.

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Appendix A – Glossary

- API – Application Programming Interface, a set of rules that allows software components to communicate with each other.
- Cloud Hosting – Deployment of applications and services on cloud infrastructure to ensure scalability and availability.
- JWT – JSON Web Token, a secure method for transmitting authentication information between client and server.
- LMS – Learning Management System, a platform used to deliver and manage educational content online.
- Online Tuition Platform – A web-based system that facilitates online teaching, learning, payments, and communication.
- Payment Gateway – A third-party service that processes online payments securely.
- RBAC – Role-Based Access Control, a security approach that restricts system access based on user roles.
- REST API – Representational State Transfer Application Programming Interface used for client-server communication.
- UI/UX – User Interface and User Experience design principles that improve usability and interaction.

Appendix B – Analysis Models

Use Case Diagram

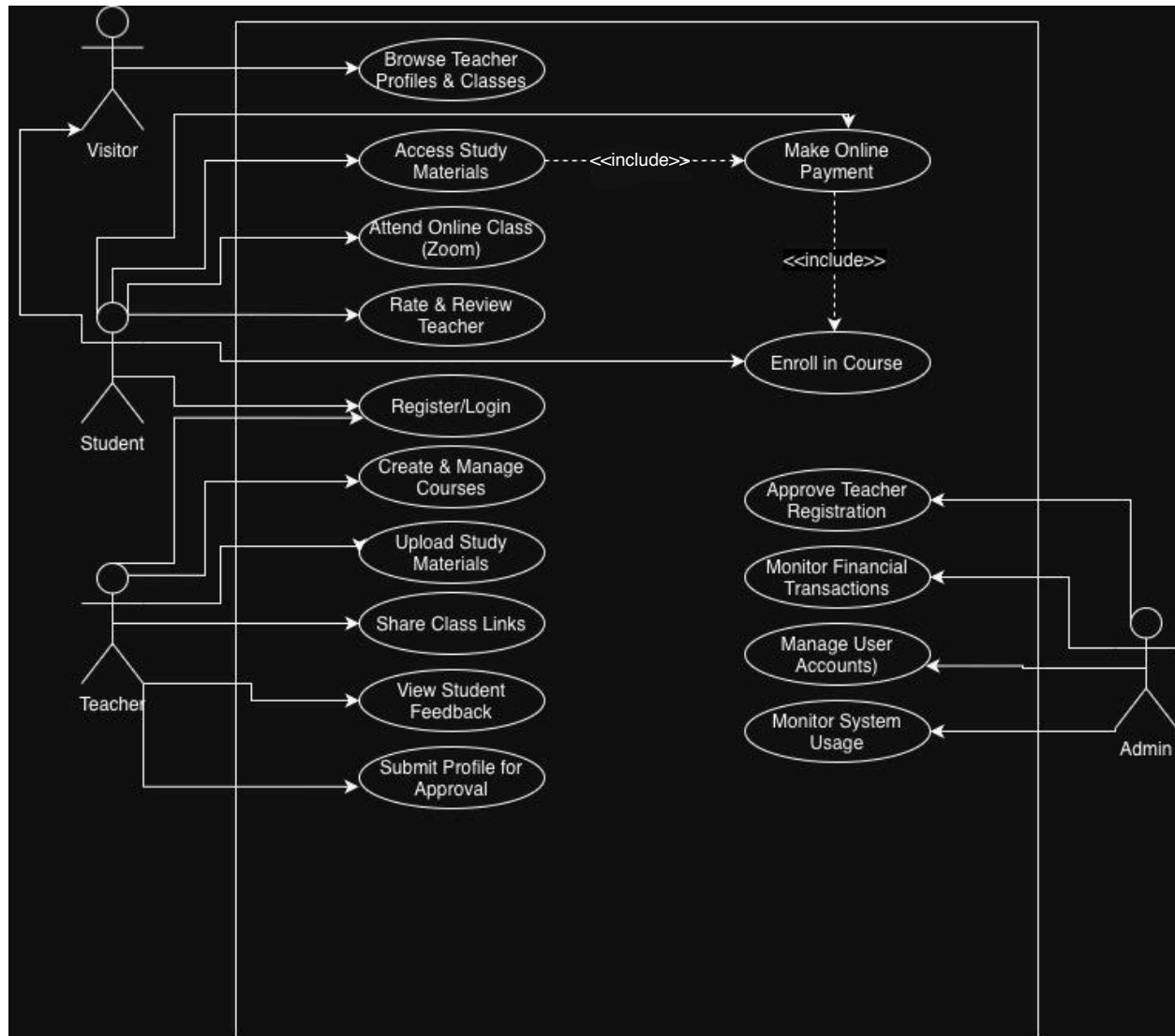


figure 8-Use Case Diagram

Class Diagram

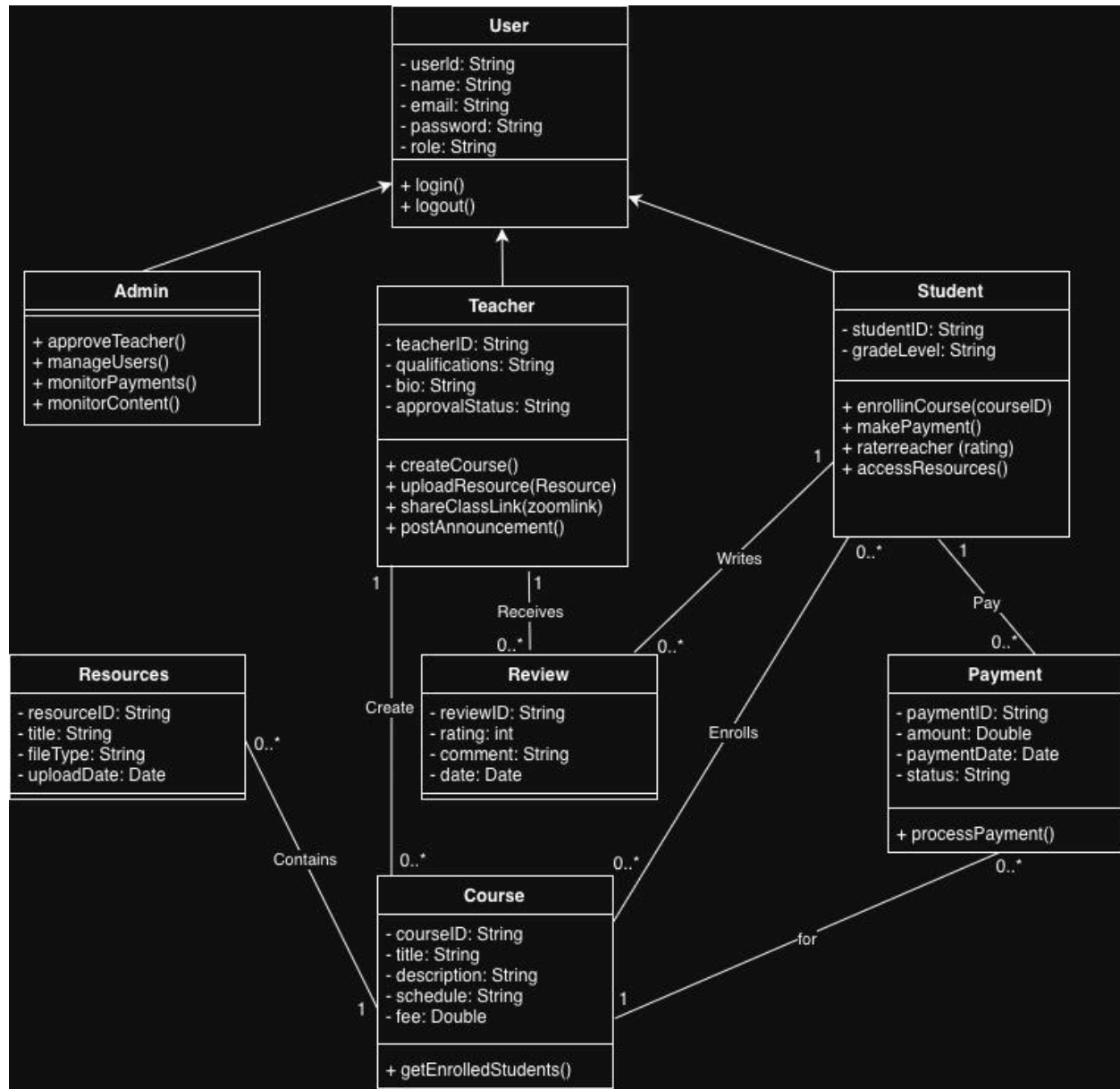


figure 9-Class Diagram

ER Diagram

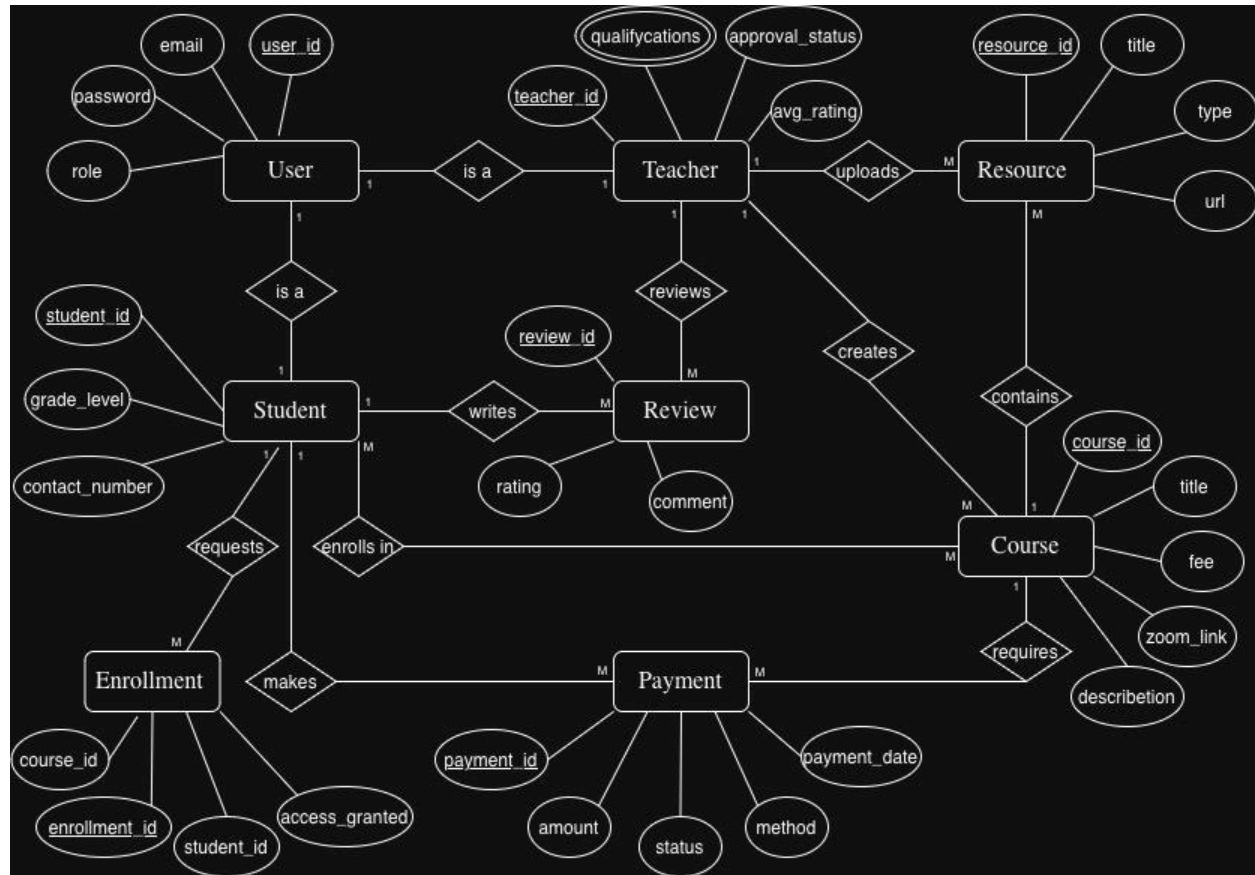


figure 10-ER Diagram

Appendix C – User Interface Designs

This appendix contains user interface mockups and screenshots that represent the visual design of the Online Tuition Class Management Platform.

Visitor Interface:

- Illustrates the public landing page displaying platform information, available teachers, and ratings.

Authentication Interfaces:

- Shows login and registration screens for students, teachers, and administrators.

Student Dashboard Interface:

- Demonstrates the dashboard layout where students can view enrolled classes, payments, materials, and notifications.

Teacher Dashboard Interface:

- Displays tools for class management, study material uploads, class link sharing, and feedback viewing.

Admin Dashboard Interface:

- Shows administrative controls for user management, teacher approval, payment monitoring, and system analytics.

Teacher Rating Interface:

- Represents the rating and review submission screen used by students.

Appendix D – Future Enhancements and Additional Notes

Future Enhancements:

- Mobile application version of the platform.
- Automated email and SMS notification system.
- AI-based teacher recommendation system.
- Advanced analytics and reporting dashboards.
- Integration with additional video conferencing tools

Additional Notes:

- The system is designed with scalability and extensibility in mind, allowing future upgrades without major architectural changes.